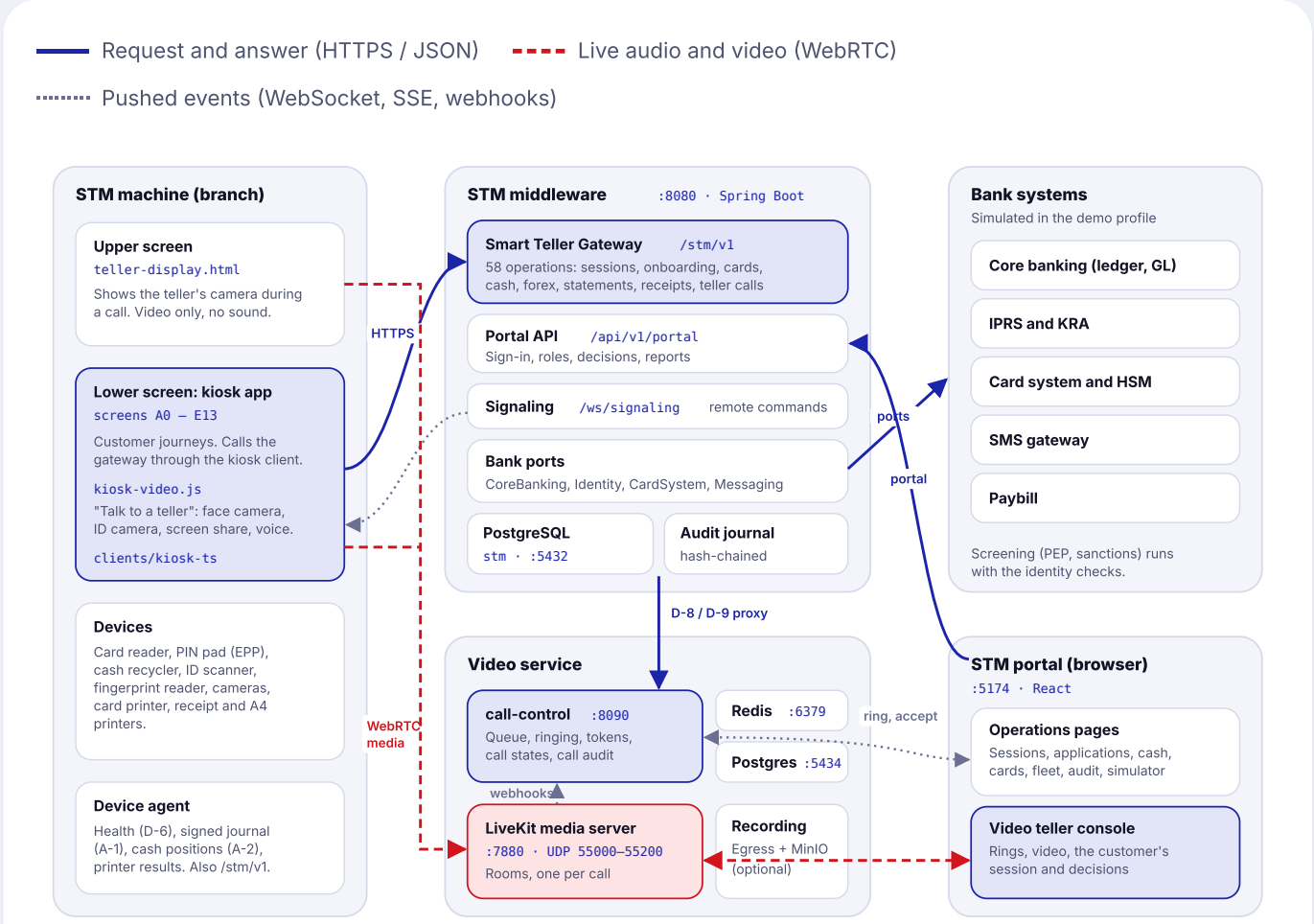
BLUE RIVER · STM KIOSK PLATFORM

How the Smart Teller works, end to end

The machine at the branch, the STM middleware that talks to the bank, the portal staff use, and the video service that connects a customer to a teller. Ports and names match your setup on the laptop.

The whole system

Two kinds of traffic leave the STM. Banking requests go to the middleware over HTTPS. Live video goes straight to LiveKit, the media server. The middleware never carries video, and LiveKit never sees account data.



`At the branch`

STM machine

The lower screen runs the kiosk app: the customer journeys from the prototype (A0 landing to E13 forex). It uses the TypeScript client to call the gateway, and `kiosk-video.js` for video help.

The upper screen runs `teller-display.html`, which shows the teller full screen during a call. A small device agent reports hardware health, uploads the machine's signed journal and reports cash at end of day.

`:8080`

STM middleware

One Spring Boot service. The **Smart Teller Gateway** (`/stm/v1`) is the only thing the machine talks to for banking. It checks who is calling, runs the business rules and limits, calls the bank through ports, and records everything in a hash-chained audit journal.

The **portal API** serves the staff portal. A **signaling** WebSocket lets a supervisor lock, unlock or reboot a kiosk remotely.

`ports`

Bank systems

Core banking, IPRS and KRA, the card system and HSM, SMS and Paybill sit behind interfaces in `backend/`. In the demo profile, simulators implement them with a double-entry ledger and test customers. For go-live, real adapters replace the simulators without changing the gateway.

`:8090 · :7880`

Video service

call-control decides who talks to whom. It queues calls, rings an available teller who speaks the customer's language, and issues short-lived LiveKit tokens. It also records the call's history.

LiveKit carries the audio and video. Each call is a private room that the kiosk, the teller and the upper screen join.

`:5174`

STM portal

The staff web app. Tellers and supervisors sign in with their own accounts. They follow customer sessions, decide referred applications, watch cash, cards and the fleet, and take video calls. Every decision is recorded under the signed-in person.

What happens inside one gateway call

Take W-2, a cash withdrawal of KES 5,000. Every `/stm/v1` request passes through the same checks before any money moves.

1. **Request ID.** The kiosk sends a fresh `X-Request-Id`. It is echoed in errors and in the audit journal, so one customer complaint can be traced end to end.

2. **Device check.** The `Authorization` device token (from D-1, valid 15 minutes) must belong to the `X-Device-Id`. In production the token is obtained over mutual TLS with the machine's certificate.

3. **Session check.** `X-Session-Id` must be an open session on this machine. A session lasts at most 30 minutes and expires after 10 minutes without activity.

4. **Customer check.** For the customer's own accounts and cards, `X-Customer-Token` is required. It comes from card + PIN (S-2), the card-services fingerprint (B-5) or a just-opened account (B-12), and only opens that customer's accounts.

5. **Business rules.** Limits (KES 40,000 per withdrawal and per day, multiples of 100), funds, and whether the recycler holds the notes. The answer to a failure is a clear message in English and Kiswahili.

6. **Idempotency.** The `Idempotency-Key` is stored before the work starts. If the kiosk retries after a timeout, it gets the first answer back instead of a second debit.

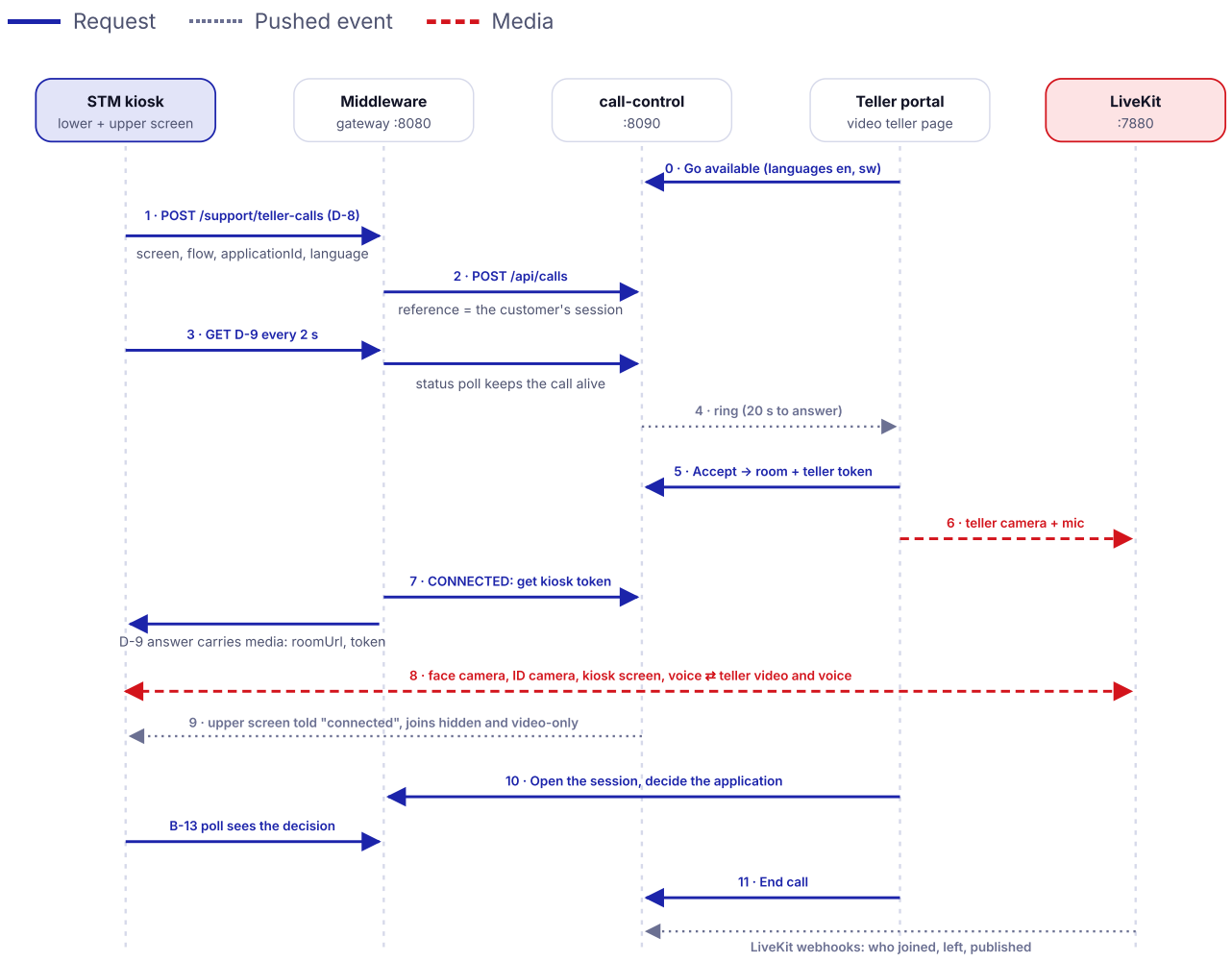
7. **Posting.** Core banking debits the account and credits the kiosk's cash GL in the same transaction (`GL-CASH-KES-<device>`). The gateway returns the note mix to dispense.

8. **Audit.** The step is written to the audit journal, linked by hash to the entry before it. The portal's Audit page re-checks the whole chain.

After dispensing, the kiosk calls W-3 if the customer took the notes, or W-4 if the shutter retracted them, which credits the customer back. At end of day the device agent reports what each cassette holds (A-2), and the Reconciliation page compares it with the GL.

A call from "Talk to a teller" to the end

The customer taps the button on any screen. The kiosk asks the gateway, so the call is tied to the customer's session. The teller then sees exactly what that customer has done so far.



1. **The teller goes available** on the portal's Video teller page, with the languages they speak. call-control keeps their presence alive with a heartbeat and pushes events to the page.

2. **The customer taps "Talk to a teller".** The kiosk calls D-8 with the screen it is on (for example `B6`), the service and any application in progress.

3. **The middleware asks call-control for a call.** The call reference is the customer's session id (`ses_…`), and the kiosk's device id becomes the call's terminal.

4. **The kiosk polls D-9 every 2 seconds.** Each poll passes through to call-control, which also tells it the customer is still waiting. If polling stops for 10 seconds, the call is dropped, so a teller never rings for a customer who has walked away.

5. **call-control rings a teller** who is available and speaks the language, first come first served. The teller has 20 seconds. A missed ring sets the teller to Away and the next teller is rung.

6. **The teller accepts.** call-control creates a LiveKit room for this call and gives the teller a two-minute token to join it. The teller's camera is published as `teller-cam`.

7. **The kiosk gets its token.** Once the call is connected, the next D-9 answer carries `media`: the room address and a token for the kiosk.

8. **The kiosk joins the room** and publishes the customer's face camera, the ID camera, its own screen and the handset microphone. The teller's voice plays on the lower screen.

9. **The upper screen** listens to call-control for its terminal. When the call connects it joins the same room hidden and receive-only, and shows only the teller's camera. It has no sound, so there is no echo, and it is never recorded.

10. **The teller helps.** The portal opens the customer's session from the call reference. If an application is waiting (a screening referral, or a face that did not match three times), the teller clears or rejects it. The kiosk sees the decision on its next B-13 poll and carries on. The customer's screen is hidden from the teller while the PIN pad is in use.

11. **The call ends.** The teller ends it, or the customer hangs up. The kiosk's next D-9 says `ENDED`, and the upper screen goes back to "Need help?". LiveKit's webhooks tell call-control who joined, left and published, so it knows when a side drops out.

> **Use the same terminal id on both screens.** Through the gateway, the call's terminal is the machine's device id, for example `STM-NBO-MOI-03`. Open the upper display with that id, `open-teller-display-on-stm.cmd 192.168.1.49 STM-NBO-MOI-03 1280 0 8090`, or it will not know when a call starts.

Where each piece runs

SERVICE	PORT	DATA IT KEEPS	STARTED WITH
STM middleware	`8080`	PostgreSQL `stm` on 5432: sessions, applications, cash, cards, audit, simulator ledger	`mvn spring-boot:run`
call-control	`8090`	Redis 6379 (queue, presence), PostgreSQL 5434 (call audit)	`start-call-control.ps1 -Port 8090`
LiveKit	`7880 · UDP 55000–55200`	Nothing kept: media only	`start-livekit.ps1 -CallControlPort 8090`
STM portal	`5174`	None: calls the middleware and call-control	`npm run dev`
Kiosk test page, upper display	`served by 8090`	None	`open-kiosk-test-on-stm.cmd`, `open-teller-display-on-stm.cmd`

Security in one paragraph

The machine proves who it is with a device token (mutual TLS in production), and every request names the session it belongs to. A customer's accounts open only after card + PIN, a

fingerprint with a one-time code, or a newly opened account. PINs exist only as encrypted PIN blocks and are never stored or logged. Money calls are idempotent, so retries cannot pay out twice. Staff sign in with their own accounts and roles, and viewing a customer's ID images is itself audited. Video rooms are private per call and joined with short-lived tokens, and the teller never sees the PIN screens.